Balancing Chemical Equations

Answer key with worked solutions

A. Understanding

1. Why balance?

> Atoms are rearranged in a reaction, never created or destroyed (**conservation of mass**), so each element must have the same number of atoms on both sides.

2. Coefficient vs subscript.

> The 2 in $2\,H_2O$ is a **coefficient**: it means two water molecules. The 2 in H_2O is a **subscript**: two H atoms in each molecule, part of what makes it water. Only coefficients may be changed; changing a subscript would change the substance (e.g. H_2O_2 is hydrogen peroxide).

B. Counting atoms

3. Count atoms.

> (a) **4 N, 12 H** (b) **3 Mg, 6 O, 6 H** (c) **4 Fe, 6 S, 24 O**

C. Balancing

4. Balance.

> (a) $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$ (b) $2\,Mg + O_2 \longrightarrow 2\,MgO$ (c) $2\,H_2O_2 \longrightarrow 2\,H_2O + O_2$ (d) $2\,Fe + 3\,Cl_2 \longrightarrow 2\,FeCl_3$

5. Balance.

> (a) $4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (b) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$ (c) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

6. Combustion.

> (a) $C_2H_4 + 3\,O_2 \longrightarrow 2\,CO_2 + 2\,H_2O$ (b) $C_2H_5OH + 3\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O$. Remember the O already in ethanol: right side 4 + 3 = 7 O, so 7 − 1 = 6 O from O_2. (c) $2\,C_4H_{10} + 13\,O_2 \longrightarrow 8\,CO_2 + 10\,H_2O$. With 1 C_4H_{10} you would need 6½ O_2, so double everything.

7. Polyatomic ions.

> (a) $2\,AgNO_3 + CaCl_2 \longrightarrow 2\,AgCl + Ca(NO_3)_2$ (b) $H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O$ (c) $Al_2(SO_4)_3 + 3\,Ca(OH)_2 \longrightarrow 2\,Al(OH)_3 + 3\,CaSO_4$

D. From words to equations

8. Sodium and water.

> $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$. Check: Na 2 = 2, H 4 = 4, O 2 = 2.

E. Challenge

9. Check and balance.

> (a) **Not balanced:** H and Cl are 1 on the left but 2 on the right. Balanced: $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$ (b) $6\,CO_2 + 6\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2$. Check: C 6 = 6, H 12 = 12, O 18 = 18.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.